

MINI PROJECT

DATA SCIENCE

COVID-19 Vaccination Dashboard for India

Dataset Overview

- Complete column descriptions with data types
- Summary statistics for all numeric columns
- Dataset metadata (records, date range, states count)

First Dose Analysis

- Total first dose count (national aggregate)
- Top N states visualization (adjustable)
- State-wise data table
- Interactive bar charts

Second Dose Analysis

- Total second dose count (national aggregate)
- Top N states visualization (adjustable)
- State-wise data table
- Completion rate tracking

Male Vaccination

- Total male vaccinations
- Top 10 states chart
- All states distribution
- Detailed data tables

Female Vaccination

- Total female vaccinations
- Top 10 states chart
- All states distribution
- Gender comparison charts
- Detailed data tables

Insights & Raw Data

- Raw dataset preview
- Automated insights generation
- Gender gap analysis
- Completion rate analysis

Dataset Information

Source

- **Dataset**: COVID-19 Vaccination Data for India
- **Source**: Kaggle - [covid19-in-india](<https://www.kaggle.com/sudalairajkumar/covid19-in-india>)
- **File**: `covid\_vaccine\_statewise.csv`

Key Columns

- `Updated On` - Date of data snapshot
- `State` - State/UT name
- `First Dose Administered` - First dose count
- `Second Dose Administered` - Second dose count
- `Male (Doses Administered)` - Male vaccination count
- `Female (Doses Administered)` - Female vaccination count
- `Total Doses Administered` - Total doses given
- And 15+ more columns for detailed analysis

Key Metrics Displayed

National Overview (KPI Cards)

- States/UTs count
- Total Doses (Millions)
- First Dose (Millions)
- Second Dose (Millions)
- Completion Rate (%)

Per Tab Metrics

- Total vaccinations by category

- Top performing states
- State-wise distributions
- Gender comparisons

Technical Stack

- **Frontend**: Streamlit
- **Data Processing**: Pandas, NumPy
- **Visualization**: Matplotlib, Seaborn
- **Styling**: Custom CSS with dark theme
- **Caching**: Streamlit cache for performance

CODE:

```
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")

st.markdown('<div class="section-title"> 🇮🇳 National Overview</div>', unsafe_allow_html=True)

k1, k2, k3, k4, k5 = st.columns(5)

with k1:
    st.metric("States / UTs", df['State'].nunique())

with k2:
    val = df[COL_T].sum() / 1e6 if has(COL_T) else 0
    st.metric("Total Doses (M)", f"{val:.1f}")

with k3:
    val = df[COL_D1].sum() / 1e6 if has(COL_D1) else 0
    st.metric("First Dose (M)", f"{val:.1f}")

with k4:
    val = df[COL_D2].sum() / 1e6 if has(COL_D2) else 0
```

```
st.metric("Second Dose (M)", f"{val:.1f}")
```

with k5:

```
if has(COL_D1) and has(COL_D2) and df[COL_D1].sum() > 0:
```

```
    pct = df[COL_D2].sum() / df[COL_D1].sum() * 100
```

```
    st.metric("Completion Rate", f"{pct:.1f}%")
```

```
else:
```

```
    st.metric("Completion Rate", "N/A")
```

```
st.markdown("<br>", unsafe_allow_html=True)
```

with tab1:

```
st.markdown('<div class="section-title"> 🇮🇳 Dataset Overview</div>', unsafe_allow_html=True)
```

```
# Dataset Description
```

```
st.markdown("### Dataset Description")
```

```
st.markdown("""
```

```
This dataset contains **COVID-19 vaccination data for India**, tracking state-wise and date-wise  
vaccination progress. The data includes information about doses administered, demographic breakdowns,  
vaccine types, and age group distributions.
```

```
""")
```

```
# Column Information
```

```
st.markdown("### Column Information")
```

```
col_info = pd.DataFrame({
```

```
    'Column Name': [
```

```
        'Updated On', 'State', 'Total Doses Administered', 'Sessions', 'Sites',
```

```
        'First Dose Administered', 'Second Dose Administered',
```

```
        'Male (Doses Administered)', 'Female (Doses Administered)', 'Transgender (Doses Administered)',
```

```
        'Covaxin (Doses Administered)', 'CoviShield (Doses Administered)', 'Sputnik V (Doses Administered)',
```

```
        'AEFI', '18-44 Years (Doses Administered)', '45-60 Years (Doses Administered)',
```

```
        '60+ Years (Doses Administered)', 'Total Individuals Vaccinated'
```

```
],
```

```
'Data Type': [
```

```
    'Date', 'String', 'Numeric', 'Numeric', 'Numeric',
```

```
    'Numeric', 'Numeric', 'Numeric', 'Numeric', 'Numeric',
```

```
    'Numeric', 'Numeric', 'Numeric', 'Numeric', 'Numeric', 'Numeric', 'Numeric', 'Numeric'
```

```

],
'Description': [
    'Date of data snapshot',
    'State or Union Territory name',
    'Cumulative total doses given',
    'Number of vaccination sessions conducted',
    'Number of vaccination sites',
    'Total first doses administered',
    'Total second doses administered',
    'Doses given to males',
    'Doses given to females',
    'Doses given to transgender individuals',
    'Doses of Covaxin vaccine',
    'Doses of CoviShield vaccine',
    'Doses of Sputnik V vaccine',
    'Adverse Events Following Immunization',
    'Doses for 18-44 age group',
    'Doses for 45-60 age group',
    'Doses for 60+ age group',
    'Total unique individuals vaccinated'
]
})

st.dataframe(col_info, use_container_width=True, height=400)

# Dataset Statistics
st.markdown("### Dataset Statistics")
c1, c2, c3, c4 = st.columns(4)
with c1:
    st.metric("Total Records", f"{len(df_raw):,}")
with c2:
    st.metric("Date Range", f"{df_raw['Updated On'].min().strftime('%d-%b-%Y')} to {df_raw['Updated On'].max().strftime('%d-%b-%Y')}")
with c3:
    st.metric("States/UTs", df['State'].nunique())
with c4:
    st.metric("Missing Values", f"{df_raw.isnull().sum().sum():,}")

```

Summary Statistics

```
st.markdown("### Summary Statistics (Latest Data per State)")
```

```
summary_cols = [COL_T, COL_D1, COL_D2, COL_M, COL_F] if all(has(c) for c in [COL_T, COL_D1, COL_D2, COL_M, COL_F]) else numeric_cols[:5]
```

```
st.dataframe(df[summary_cols].describe().T.style.format("{:.0f}"), use_container_width=True)
```



